

Class 10 Math — Medium Practice Set

100 Questions | Level: Medium

StudyShelf — studysshelf practice worksheet. Answer key provided at the end.

- Q1.** Find the volume of a cone with radius 13 cm and height 11 cm. (use $\pi = 22/7$)
- Q2.** Find the area of the triangle whose vertices are A(3,10), B(6,0) and C(6,7).
- Q3.** Find the median of the data: 13, 16, 21, 32, 34, 40, 41
- Q4.** Triangle ABC ~ Triangle PQR with a scale factor of 3. If AB = 10 cm, find PQ.
- Q5.** Find the sum of the first 14 terms of the AP with first term 8 and common difference 3.
- Q6.** Find the median of the data: 3, 16, 17, 29, 31, 32, 34, 43, 46, 46, 49
- Q7.** The angle of elevation of the top of a tower from a point 20 m away from its base is 30° . Find the height of the tower.
- Q8.** Triangle ABC ~ Triangle PQR with a scale factor of 2. If AB = 4 cm, find PQ.
- Q9.** Find the volume of a cylinder with radius 7 cm and height 10 cm. (use $\pi = 22/7$)
- Q10.** Find the area of the triangle whose vertices are A(3,1), B(5,2) and C(3,10).
- Q11.** Find the area of the triangle whose vertices are A(6,3), B(4,10) and C(5,3).
- Q12.** The angle of elevation of the top of a tower from a point 30 m away from its base is 45° . Find the height of the tower.
- Q13.** Find the median of the data: 4, 13, 23, 25, 26, 40, 50
- Q14.** Find the mode of the data: 6, 14, 6, 6, 12, 19, 12, 8
- Q15.** Triangle ABC ~ Triangle PQR with a scale factor of 2.5. If AB = 13 cm, find PQ.
- Q16.** Find the sum of the first 19 terms of the AP with first term 2 and common difference 4.
- Q17.** Find the coordinates of the point which divides the line segment joining A(4,9) and B(10,7) in the ratio 3:2 internally.
- Q18.** Find the mode of the data: 18, 6, 6, 15, 5, 2, 6, 15
- Q19.** Solve the pair of linear equations: $2x + 5y = 26$ and $2x + 1y = 18$
- Q20.** Solve the pair of linear equations: $4x + 1y = 16$ and $3x + 2y = 22$
- Q21.** Find the mode of the data: 2, 3, 14, 14, 18, 7, 15, 14
- Q22.** Using the quadratic formula, solve: $x^2 - 14x + 45 = 0$
- Q23.** Find the median of the data: 2, 3, 4, 6, 14, 15, 21, 27, 33, 36, 42
- Q24.** Triangle ABC ~ Triangle PQR with a scale factor of 2.5. If AB = 10 cm, find PQ.

- Q25.** A bag contains 35 balls, out of which 25 are red and the rest are blue. A ball is drawn at random. Find the probability that it is red.
- Q26.** Find the area of the triangle whose vertices are A(1,9), B(8,8) and C(8,7).
- Q27.** Find the median of the data: 8, 16, 17, 20, 32, 35, 36, 41, 43
- Q28.** Find the mode of the data: 20, 9, 8, 8, 12, 4, 20, 8
- Q29.** Find the sum of the first 17 terms of the AP with first term 4 and common difference 6.
- Q30.** Find the area of the triangle whose vertices are A(3,1), B(5,10) and C(4,10).
- Q31.** Find the area of the triangle whose vertices are A(3,4), B(3,2) and C(7,0).
- Q32.** Solve the pair of linear equations: $3x + 2y = 28$ and $2x + 2y = 22$
- Q33.** Find the volume of a cylinder with radius 3 cm and height 6 cm. (use $\pi = 22/7$)
- Q34.** The angle of elevation of the top of a tower from a point 15 m away from its base is 60° . Find the height of the tower.
- Q35.** Find the sum of the first 9 terms of the AP with first term 3 and common difference 6.
- Q36.** Find the volume of a sphere of radius 3 cm. (use $\pi = 22/7$)
- Q37.** A bag contains 34 balls, out of which 13 are red and the rest are blue. A ball is drawn at random. Find the probability that it is red.
- Q38.** Triangle ABC ~ Triangle PQR with a scale factor of 3. If AB = 8 cm, find PQ.
- Q39.** Find the sum of the first 12 terms of the AP with first term 3 and common difference 3.
- Q40.** Triangle ABC ~ Triangle PQR with a scale factor of 1.5. If AB = 11 cm, find PQ.
- Q41.** Find the area of the triangle whose vertices are A(0,10), B(8,7) and C(3,3).
- Q42.** Find the mode of the data: 1, 3, 17, 17, 12, 17, 11, 7
- Q43.** Find the coordinates of the point which divides the line segment joining A(7,9) and B(6,5) in the ratio 4:4 internally.
- Q44.** A bag contains 28 balls, out of which 19 are red and the rest are blue. A ball is drawn at random. Find the probability that it is red.
- Q45.** Find the mode of the data: 8, 8, 8, 12, 3, 7, 8, 3
- Q46.** Find the median of the data: 1, 4, 5, 5, 6, 32, 49
- Q47.** Find the sum of the first 16 terms of the AP with first term 7 and common difference 4.
- Q48.** The angle of elevation of the top of a tower from a point 30 m away from its base is 30° . Find the height of the tower.
- Q49.** Solve the pair of linear equations: $1x + 2y = 14$ and $3x + 5y = 38$
- Q50.** Find the volume of a cylinder with radius 11 cm and height 15 cm. (use $\pi = 22/7$)

- Q51.** Find the coordinates of the point which divides the line segment joining A(3,5) and B(10,10) in the ratio 4:1 internally.
- Q52.** Find the mode of the data: 5, 1, 5, 4, 8, 5, 20, 5
- Q53.** The angle of elevation of the top of a tower from a point 10 m away from its base is 30° . Find the height of the tower.
- Q54.** Find the sum of the first 11 terms of the AP with first term 6 and common difference 6.
- Q55.** Find the sum of the first 15 terms of the AP with first term 8 and common difference 2.
- Q56.** Solve the pair of linear equations: $5x + 4y = 85$ and $4x + 4y = 76$
- Q57.** Find the coordinates of the point which divides the line segment joining A(10,3) and B(2,6) in the ratio 3:2 internally.
- Q58.** Find the mode of the data: 3, 12, 14, 9, 14, 14, 20, 2
- Q59.** Using the quadratic formula, solve: $x^2 - 17x + 70 = 0$
- Q60.** Solve the pair of linear equations: $5x + 3y = 16$ and $4x + 3y = 14$
- Q61.** Find the volume of a cylinder with radius 11 cm and height 7 cm. (use $\pi = 22/7$)
- Q62.** Find the volume of a cone with radius 13 cm and height 5 cm. (use $\pi = 22/7$)
- Q63.** Solve the pair of linear equations: $2x + 4y = 46$ and $2x + 1y = 25$
- Q64.** A bag contains 24 balls, out of which 2 are red and the rest are blue. A ball is drawn at random. Find the probability that it is red.
- Q65.** Find the mode of the data: 11, 11, 11, 14, 13, 6, 11, 2
- Q66.** Find the coordinates of the point which divides the line segment joining A(4,8) and B(1,3) in the ratio 3:4 internally.
- Q67.** Find the area of the triangle whose vertices are A(1,4), B(10,2) and C(1,7).
- Q68.** Find the mode of the data: 15, 18, 10, 2, 13, 10, 10, 15
- Q69.** Solve the pair of linear equations: $4x + 5y = 54$ and $2x + 3y = 32$
- Q70.** Find the median of the data: 8, 20, 28, 31, 31, 33, 35, 39, 40
- Q71.** Solve the pair of linear equations: $2x + 3y = 20$ and $4x + 4y = 36$
- Q72.** Find the volume of a sphere of radius 8 cm. (use $\pi = 22/7$)
- Q73.** Find the sum of the first 8 terms of the AP with first term 3 and common difference 5.
- Q74.** Find the sum of the first 19 terms of the AP with first term 3 and common difference 5.
- Q75.** A bag contains 23 balls, out of which 20 are red and the rest are blue. A ball is drawn at random. Find the probability that it is red.
- Q76.** Using the quadratic formula, solve: $x^2 - 17x + 66 = 0$

Q77. Find the median of the data: 4, 4, 14, 23, 42, 49, 50

Q78. Find the volume of a sphere of radius 12 cm. (use $\pi = 22/7$)

Q79. Find the area of the triangle whose vertices are A(10,6), B(1,2) and C(2,9).

Q80. The angle of elevation of the top of a tower from a point 25 m away from its base is 30° . Find the height of the tower.

Q81. A bag contains 25 balls, out of which 21 are red and the rest are blue. A ball is drawn at random. Find the probability that it is red.

Q82. A bag contains 35 balls, out of which 29 are red and the rest are blue. A ball is drawn at random. Find the probability that it is red.

Q83. Find the coordinates of the point which divides the line segment joining A(8,6) and B(7,6) in the ratio 2:1 internally.

Q84. The angle of elevation of the top of a tower from a point 25 m away from its base is 45° . Find the height of the tower.

Q85. Find the median of the data: 6, 20, 25, 26, 33, 38, 45

Q86. Find the volume of a sphere of radius 14 cm. (use $\pi = 22/7$)

Q87. Solve the pair of linear equations: $5x + 5y = 60$ and $4x + 5y = 58$

Q88. A bag contains 25 balls, out of which 2 are red and the rest are blue. A ball is drawn at random. Find the probability that it is red.

Q89. Find the median of the data: 8, 12, 29, 29, 32, 33, 37, 42, 45, 45, 49

Q90. Using the quadratic formula, solve: $x^2 - 10x + 16 = 0$

Q91. Find the area of the triangle whose vertices are A(7,4), B(8,9) and C(5,10).

Q92. Find the sum of the first 17 terms of the AP with first term 4 and common difference 4.

Q93. Find the area of the triangle whose vertices are A(4,3), B(3,5) and C(2,6).

Q94. Find the volume of a sphere of radius 5 cm. (use $\pi = 22/7$)

Q95. Using the quadratic formula, solve: $x^2 - 15x + 44 = 0$

Q96. A bag contains 32 balls, out of which 4 are red and the rest are blue. A ball is drawn at random. Find the probability that it is red.

Q97. Find the volume of a cylinder with radius 8 cm and height 17 cm. (use $\pi = 22/7$)

Q98. Using the quadratic formula, solve: $x^2 - 10x + 21 = 0$

Q99. Find the sum of the first 16 terms of the AP with first term 5 and common difference 3.

Q100. Find the area of the triangle whose vertices are A(2,10), B(10,10) and C(4,9).

Answer Key

1. Volume $\approx 1947.52 \text{ cm}^3$
2. Area = 10.5 sq. units
3. Median = 32
4. PQ = 30 cm
5. Sum = 385
6. Median = 32
7. Height $\approx 11.55 \text{ m}$
8. PQ = 8 cm
9. Volume $\approx 1540.0 \text{ cm}^3$
10. Area = 9.0 sq. units
11. Area = 3.5 sq. units
12. Height $\approx 30.00 \text{ m}$
13. Median = 25
14. Mode = 6
15. PQ = 32.5 cm
16. Sum = 722
17. Point = $(38/5, 39/5)$
18. Mode = 6
19. $x = 8, y = 2$
20. $x = 2, y = 8$
21. Mode = 14
22. $x = 5$ or $x = 9$
23. Median = 15
24. PQ = 25.0 cm
25. $P(\text{red}) = 25/35 = 5/7$
26. Area = 3.5 sq. units
27. Median = 32
28. Mode = 8
29. Sum = 884
30. Area = 4.5 sq. units
31. Area = 4.0 sq. units
32. $x = 6, y = 5$
33. Volume $\approx 169.71 \text{ cm}^3$
34. Height $\approx 25.98 \text{ m}$
35. Sum = 243
36. Volume $\approx 113.14 \text{ cm}^3$
37. $P(\text{red}) = 13/34 = 13/34$

38. $PQ = 24 \text{ cm}$
39. $\text{Sum} = 234$
40. $PQ = 16.5 \text{ cm}$
41. $\text{Area} = 23.5 \text{ sq. units}$
42. $\text{Mode} = 17$
43. $\text{Point} = (13/2, 7)$
44. $P(\text{red}) = 19/28 = 19/28$
45. $\text{Mode} = 8$
46. $\text{Median} = 5$
47. $\text{Sum} = 592$
48. $\text{Height} \approx 17.32 \text{ m}$
49. $x = 6, y = 4$
50. $\text{Volume} \approx 5704.29 \text{ cm}^3$
51. $\text{Point} = (43/5, 9)$
52. $\text{Mode} = 5$
53. $\text{Height} \approx 5.77 \text{ m}$
54. $\text{Sum} = 396$
55. $\text{Sum} = 330$
56. $x = 9, y = 10$
57. $\text{Point} = (26/5, 24/5)$
58. $\text{Mode} = 14$
59. $x = 7 \text{ or } x = 10$
60. $x = 2, y = 2$
61. $\text{Volume} \approx 2662.0 \text{ cm}^3$
62. $\text{Volume} \approx 885.24 \text{ cm}^3$
63. $x = 9, y = 7$
64. $P(\text{red}) = 2/24 = 1/12$
65. $\text{Mode} = 11$
66. $\text{Point} = (19/7, 41/7)$
67. $\text{Area} = 13.5 \text{ sq. units}$
68. $\text{Mode} = 10$
69. $x = 1, y = 10$
70. $\text{Median} = 31$
71. $x = 7, y = 2$
72. $\text{Volume} \approx 2145.52 \text{ cm}^3$
73. $\text{Sum} = 164$
74. $\text{Sum} = 912$
75. $P(\text{red}) = 20/23 = 20/23$
76. $x = 6 \text{ or } x = 11$

77. Median = 23

78. Volume $\approx 7241.14 \text{ cm}^3$

79. Area = 29.5 sq. units

80. Height $\approx 14.43 \text{ m}$

81. $P(\text{red}) = 21/25 = 21/25$

82. $P(\text{red}) = 29/35 = 29/35$

83. Point = $(22/3, 6)$

84. Height $\approx 25.00 \text{ m}$

85. Median = 26

86. Volume $\approx 11498.67 \text{ cm}^3$

87. $x = 2, y = 10$

88. $P(\text{red}) = 2/25 = 2/25$

89. Median = 33

90. $x = 8$ or $x = 2$

91. Area = 8.0 sq. units

92. Sum = 612

93. Area = 0.5 sq. units

94. Volume $\approx 523.81 \text{ cm}^3$

95. $x = 11$ or $x = 4$

96. $P(\text{red}) = 4/32 = 1/8$

97. Volume $\approx 3419.43 \text{ cm}^3$

98. $x = 3$ or $x = 7$

99. Sum = 440

100. Area = 4.0 sq. units